

5.5.4

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

Find the value of k for which the matrix is singular.

$$\begin{pmatrix} k & 8 \\ 1 & 2k \end{pmatrix}$$

Solution:

Given

$$\mathbf{A} = \begin{pmatrix} k & 8 \\ 1 & 2k \end{pmatrix} \quad (1)$$

$$R_1 \leftrightarrow R_2 \Rightarrow \begin{pmatrix} 1 & 2k \\ k & 8 \end{pmatrix} \quad (2)$$

$$R_2 \rightarrow R_2 - kR_1 \Rightarrow \begin{pmatrix} 1 & 2k \\ 0 & 8 - 2k^2 \end{pmatrix} \quad (3)$$

For $\mathbf{A}$ to be singular, the second row must become zero because it should not be a full rank matrix:

$$8 - 2k^2 = 0 \quad (4)$$

$$k^2 = 4 \quad (5)$$

$$k = \pm 2 \quad (6)$$